

## HOORN-A, Netherlands

WMO: 062120

Lat: 52.918N

Lon: 4.150E

Elev: 41

StdP: 100.83

Time Zone: 1.00 (EUC)

Period: 08-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -1.2       | 0.0 | -7.0                          | 2.1 | -0.2 | -5.2 | 2.4 | 0.6  | 20.4                  | 8.2  | 19.0 | 8.0  | 10.5                  | 80   | 0.829 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 2.2                    | 22.1            | 18.6 | 20.6 | 17.9 | 19.5 | 17.2 | 19.2                | 21.6 | 18.4 | 20.2 | 17.6 | 19.2 | 6.7                  | 110  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.3                            | 13.3 | 20.5 | 17.5 | 12.6 | 19.6 | 16.8 | 12.0 | 18.9 | 55.0          | 21.7 | 52.3 | 20.4 | 49.8 | 19.1 | 24.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |    |    |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|----|----|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |    |    |
| (4) 17.9          | 16.3 | 14.7 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |    |    |
| (5)               |      |      | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 11.0        | 5.7   | 5.3   | 6.2   | 8.8   | 11.3  | 14.1  | 16.7  | 17.4  | 16.1  | 13.5  | 9.7   | 7.4   |  |  |
|                                                                        | DBStd       | 4.87        | 2.55  | 2.37  | 2.43  | 2.56  | 2.53  | 2.12  | 2.11  | 1.88  | 1.92  | 2.13  | 2.51  | 2.76  |  |  |
|                                                                        | HDD10.0     | 570         | 133   | 132   | 120   | 51    | 13    | 0     | 0     | 0     | 0     | 2     | 34    | 85    |  |  |
|                                                                        | HDD18.3     | 2686        | 392   | 365   | 377   | 285   | 218   | 131   | 59    | 38    | 71    | 151   | 259   | 339   |  |  |
|                                                                        | CDD10.0     | 953         | 0     | 0     | 1     | 16    | 54    | 122   | 208   | 231   | 184   | 109   | 24    | 4     |  |  |
|                                                                        | CDD18.3     | 26          | 0     | 0     | 0     | 0     | 0     | 2     | 9     | 11    | 4     | 0     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 38          | 0     | 0     | 0     | 0     | 0     | 5     | 21    | 7     | 6     | 0     | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 4           | 0     | 0     | 0     | 0     | 0     | 0     | 4     | 0     | 0     | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 7.9         | 9.4   | 9.1   | 8.3   | 7.1   | 7.2   | 6.6   | 6.3   | 6.6   | 7.1   | 8.5   | 8.7   | 9.6   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 902         | 76    | 63    | 52    | 43    | 55    | 66    | 81    | 99    | 90    | 94    | 96    | 87    |  |  |
|                                                                        | PrecMax     | 1188        | 141   | 130   | 118   | 100   | 99    | 152   | 173   | 227   | 196   | 178   | 175   | 158   |  |  |
|                                                                        | PrecMin     | 661         | 7     | 21    | 10    | 0     | 21    | 25    | 20    | 22    | 14    | 21    | 10    | 22    |  |  |
|                                                                        | PrecStd     | 126         | 35    | 29    | 28    | 24    | 23    | 35    | 41    | 55    | 46    | 41    | 39    | 35    |  |  |
| (20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 10.7  | 10.7  | 12.7  | 17.9  | 20.4  | 23.6  | 26.5  | 24.4  | 23.9  | 18.8  | 15.7  | 12.6  |  |  |
|                                                                        |             | MCWB        | 9.4   | 7.9   | 9.5   | 12.2  | 14.2  | 18.6  | 20.1  | 19.4  | 19.9  | 16.4  | 14.2  | 11.2  |  |  |
|                                                                        | 2%          | DB          | 10.1  | 9.5   | 10.5  | 15.3  | 17.3  | 19.4  | 21.7  | 21.7  | 20.5  | 17.6  | 14.7  | 12.1  |  |  |
|                                                                        |             | MCWB        | 8.8   | 7.6   | 8.5   | 11.4  | 13.7  | 17.1  | 18.5  | 18.3  | 18.2  | 15.6  | 13.2  | 10.8  |  |  |
|                                                                        | 5%          | DB          | 9.5   | 8.8   | 9.5   | 13.2  | 15.6  | 17.5  | 20.3  | 20.4  | 19.3  | 16.8  | 14.0  | 11.4  |  |  |
|                                                                        |             | MCWB        | 8.2   | 7.3   | 7.9   | 10.3  | 13.2  | 15.7  | 17.9  | 17.8  | 17.2  | 14.9  | 12.5  | 10.1  |  |  |
|                                                                        | 10%         | DB          | 8.8   | 8.2   | 8.7   | 11.8  | 14.3  | 16.3  | 19.0  | 19.4  | 18.4  | 16.2  | 13.1  | 10.6  |  |  |
|                                                                        |             | MCWB        | 7.5   | 6.8   | 7.4   | 9.8   | 12.6  | 14.7  | 16.9  | 17.1  | 16.3  | 14.4  | 11.2  | 9.2   |  |  |
| (28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 9.9   | 9.2   | 10.0  | 12.8  | 15.6  | 18.8  | 20.2  | 19.9  | 20.6  | 16.7  | 14.7  | 11.7  |  |  |
|                                                                        |             | MCDB        | 10.5  | 9.8   | 12.3  | 16.8  | 18.1  | 21.8  | 23.4  | 22.8  | 23.4  | 18.2  | 15.4  | 12.3  |  |  |
|                                                                        | 2%          | WB          | 9.1   | 8.1   | 9.0   | 11.6  | 14.3  | 17.4  | 19.1  | 19.0  | 19.0  | 15.9  | 13.6  | 11.1  |  |  |
|                                                                        |             | MCDB        | 9.9   | 8.9   | 10.1  | 14.5  | 16.3  | 19.3  | 21.6  | 21.1  | 20.7  | 17.2  | 14.6  | 11.9  |  |  |
|                                                                        | 5%          | WB          | 8.4   | 7.5   | 8.3   | 10.7  | 13.5  | 16.0  | 18.2  | 18.2  | 17.6  | 15.3  | 12.7  | 10.4  |  |  |
|                                                                        |             | MCDB        | 9.3   | 8.5   | 9.3   | 12.6  | 15.3  | 17.3  | 20.0  | 19.8  | 18.9  | 16.6  | 13.9  | 11.3  |  |  |
|                                                                        | 10%         | WB          | 7.8   | 7.0   | 7.6   | 10.0  | 12.7  | 14.9  | 17.2  | 17.6  | 16.8  | 14.7  | 11.4  | 9.4   |  |  |
|                                                                        |             | MCDB        | 8.8   | 8.1   | 8.6   | 11.6  | 14.4  | 16.0  | 18.6  | 19.2  | 18.3  | 16.1  | 12.8  | 10.5  |  |  |
| (36) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 2.4   | 2.1   | 2.4   | 2.7   | 2.4   | 2.2   | 2.3   | 2.2   | 2.1   | 2.4   | 2.4   | 2.4   |  |  |
|                                                                        |             | MCDBR       | 2.6   | 2.4   | 3.3   | 5.0   | 4.9   | 4.7   | 4.5   | 3.9   | 2.9   | 2.7   | 2.9   | 2.2   |  |  |
|                                                                        | 5% WB       | MCWBR       | 3.0   | 2.2   | 2.5   | 2.9   | 2.9   | 3.3   | 3.0   | 2.6   | 2.8   | 2.7   | 3.2   | 2.9   |  |  |
|                                                                        |             | MCWBR       | 2.6   | 2.2   | 3.1   | 4.5   | 4.1   | 4.6   | 4.2   | 3.3   | 2.7   | 2.7   | 2.8   | 2.2   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.334 | 0.350 | 0.379 | 0.402 | 0.402 | 0.401 | 0.409 | 0.401 | 0.385 | 0.366 | 0.343 | 0.325 |  |  |
|                                                                        |             | taud        | 2.327 | 2.355 | 2.304 | 2.285 | 2.354 | 2.369 | 2.356 | 2.391 | 2.426 | 2.414 | 2.360 | 2.320 |  |  |
|                                                                        | Ebn at Noon |             | 636   | 749   | 800   | 829   | 849   | 852   | 837   | 823   | 790   | 721   | 619   | 571   |  |  |
|                                                                        |             | Edh at Noon | 62    | 80    | 102   | 118   | 116   | 115   | 115   | 105   | 91    | 75    | 60    | 53    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 0.64  | 1.33  | 2.64  | 4.33  | 5.35  | 5.62  | 5.38  | 4.50  | 3.15  | 1.73  | 0.76  | 0.48  |  |  |
|                                                                        | RadStd      |             | 0.06  | 0.23  | 0.25  | 0.40  | 0.33  | 0.33  | 0.54  | 0.33  | 0.25  | 0.15  | 0.07  | 0.05  |  |  |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (57 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page